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Project Title	Forecasting Repeat Customer Purchases Using BG/NBD
Week	Week 1

WEEK-1 Technical Report

Task Name

Forecasting Repeat Customer Purchases Using BG/NBD

Abstract

Customer retention is one of the most important factors for business growth and profitability. Predicting future customer purchases helps companies design effective marketing strategies and improve customer engagement. This project implements the BG/NBD (Beta Geometric/Negative Binomial Distribution) model to forecast repeat customer purchases based on historical transaction data. The model estimates customer purchasing behavior and predicts the likelihood of future purchases. By analyzing customer transaction patterns, businesses can identify valuable customers, improve retention strategies, and increase revenue.

Introduction

In modern business environments, understanding customer behavior is essential for maintaining long-term relationships and maximizing profits. Customer Lifetime Value (CLV) analysis helps organizations identify customers who are likely to make future purchases. The BG/NBD model is a probabilistic model widely used in customer analytics to forecast repeat buying behavior. This project focuses on applying the BG/NBD model to predict future purchases and support data-driven marketing decisions.

Theory

The BG/NBD model is based on the following concepts:

- Customer transactions follow a Poisson process.
- Customer dropout probability follows a geometric distribution.
- Purchase frequency and recency are key factors in prediction.
- Probabilistic modeling estimates future purchasing behavior.
- Customer Lifetime Value can be derived from purchase forecasts.

Objectives

- Understand the BG/NBD customer behavior model.
- Analyze customer transaction history.
- Calculate purchase frequency and recency.
- Forecast future customer purchases.
- Identify high-value customers.
- Support customer retention and marketing strategies.

Methodology

Step 1: Data Collection

Collect customer transaction data including purchase dates, customer IDs, and transaction amounts.

Step 2: Data Preprocessing

Clean transaction records, remove duplicates, handle missing values, and prepare customer-level metrics.

Step 3: Feature Extraction

Calculate Recency, Frequency, and Customer Age (T) from transaction history.

Step 4: Model Development

Apply the BG/NBD model to estimate purchasing behavior and customer retention probabilities.

Step 5: Forecasting and Evaluation

Predict future purchases and evaluate model performance using historical transaction patterns.

Dataset

The dataset contains:

- Customer ID
- Transaction Date
- Purchase Frequency
- Recency
- Customer Age (T)
- Transaction Amount
- Purchase History
- Future Purchase Indicator

Mathematical Model

The BG/NBD model estimates the probability that a customer will make future purchases based on:

- Frequency (number of repeat purchases)

- Recency (time since last purchase)
- Customer Lifetime (T)

The model combines:

- Beta Distribution for customer dropout behavior.
- Negative Binomial Distribution for purchase frequency.

These probabilistic distributions help forecast future transaction counts.

Complexity Analysis

- Training Complexity: $O(n)$
- Forecasting Complexity: $O(n)$
- Memory Complexity: $O(n)$

where n represents the number of customers in the dataset.

Results

The model successfully:

- Predicts future customer purchases.
- Identifies active and inactive customers.
- Estimates customer retention probability.
- Supports targeted marketing campaigns.
- Improves customer relationship management.

Applications

- Customer Relationship Management (CRM)
- Customer Lifetime Value Analysis

- Marketing Campaign Optimization
- E-commerce Analytics
- Retail Business Intelligence
- Customer Retention Strategies

Conclusion

This project demonstrates the application of the BG/NBD model for forecasting repeat customer purchases. By analyzing customer transaction history and estimating future buying behavior, the model helps businesses identify valuable customers and improve retention strategies. The approach provides meaningful insights for customer analytics, marketing optimization, and long-term business growth.